

A Universal World Model Learned from Large Scale and Diverse Videos

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1. Why training a universal world model using video is important?

- Low generalization ability of current world models
- A massive amount of state-action pairs is required to train a world model
- Video contains rich semantic and dynamic information
- Video is abundant and large scale compared to environment-generated samples

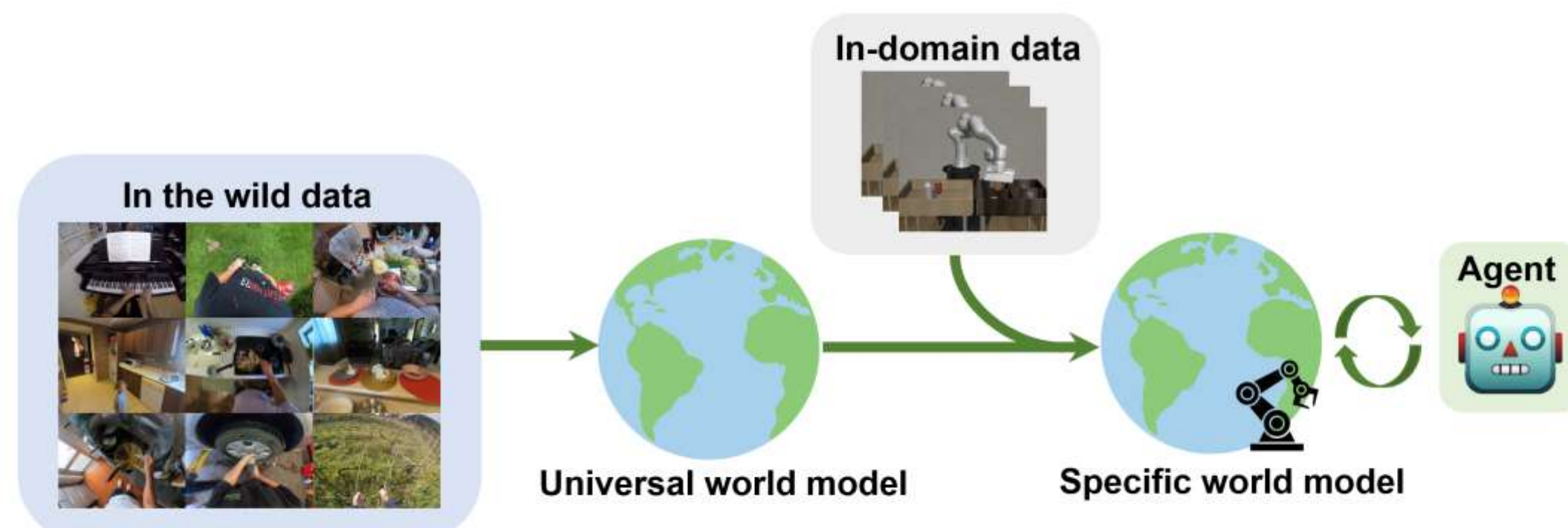


Figure 1: Pipeline of the proposed universal world model

2. Why it is challenge to learn a universal world model from video?

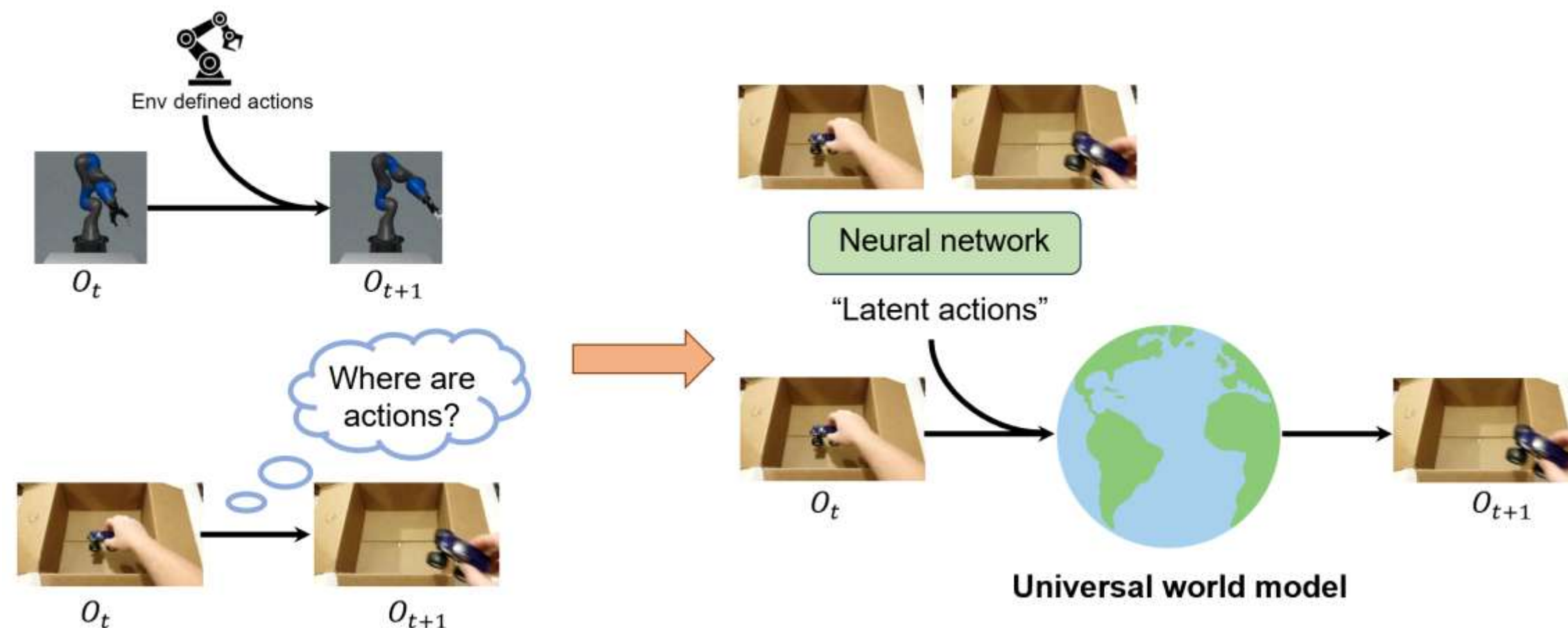


Figure 2: How to define actions in action-free videos

3. How to efficiently align the learned actions with env-specific actions?

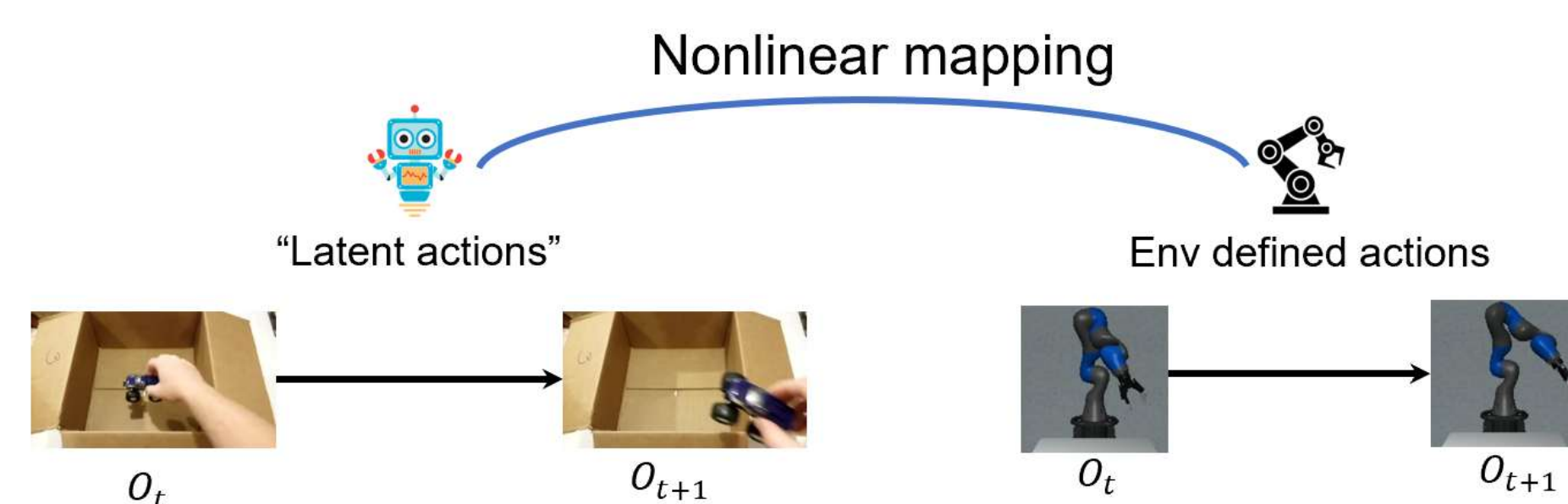


Figure 3: Align learned actions and real actions by finetuning

4. How to extract latent actions and align them to real actions?

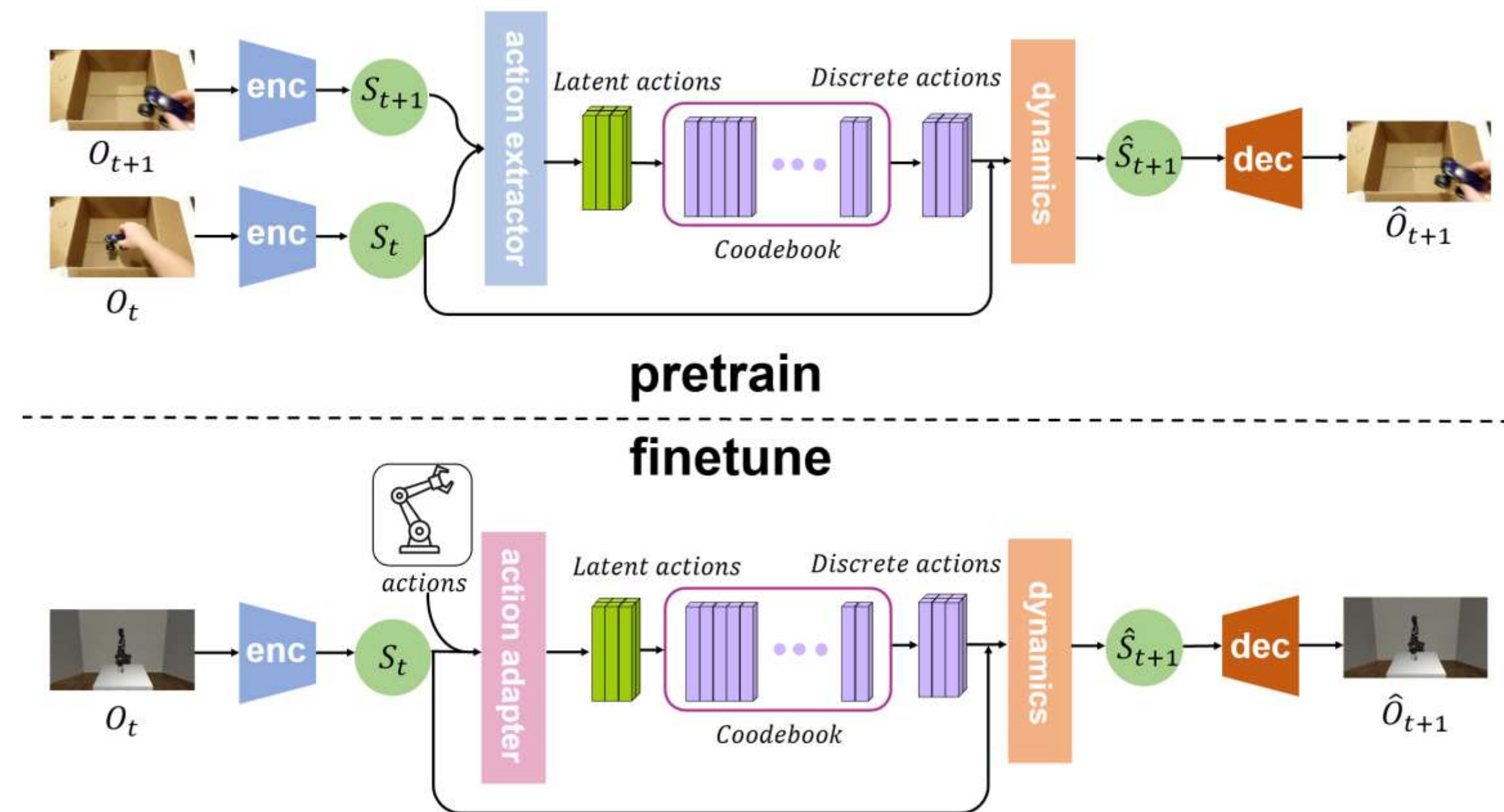


Figure 4: The general framework of the proposed universal world model

5. Visualization results

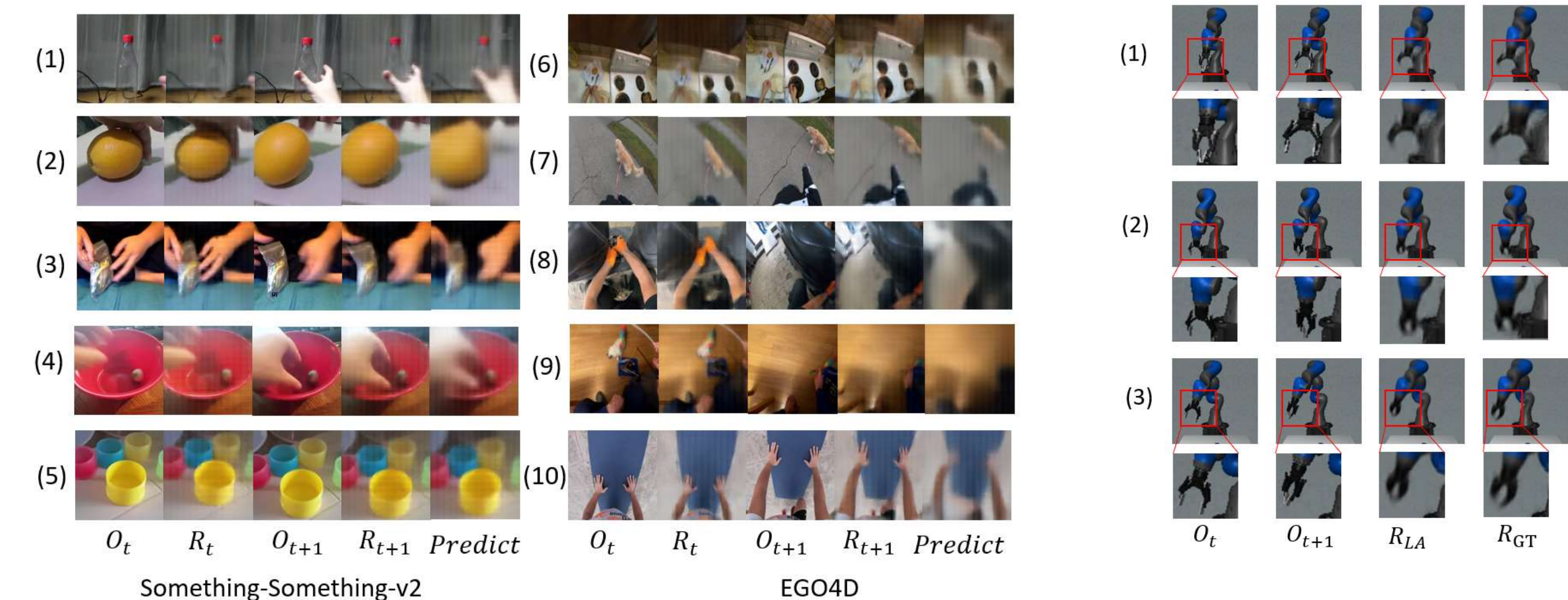


Figure 5: Predicted frames of in-the-wild videos and domain-specific videos

6. Metrics and quantitative evaluation

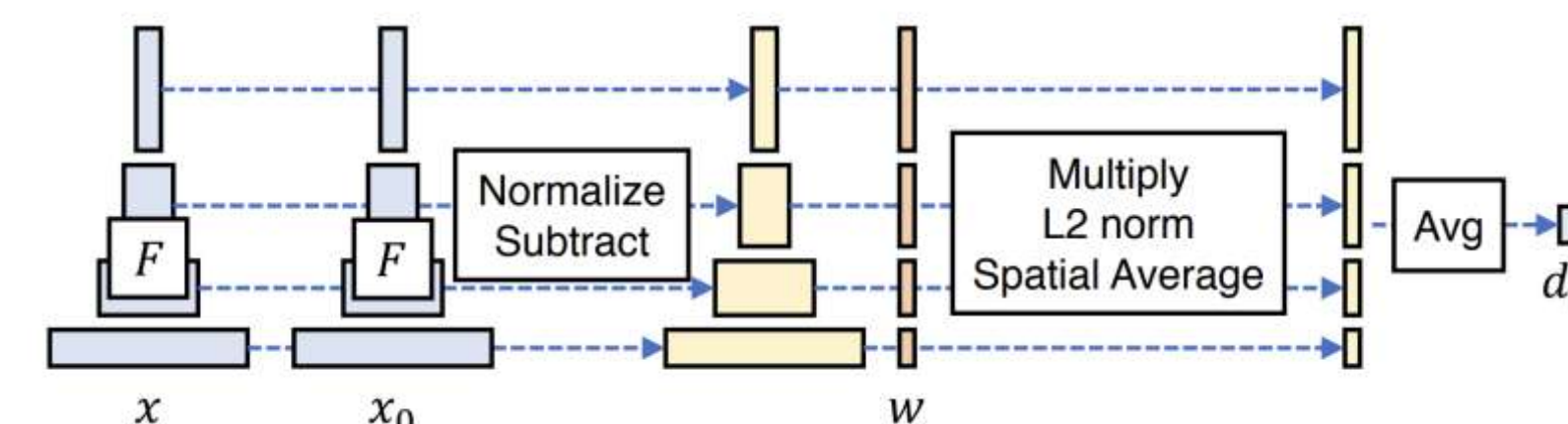


Figure 6: Learned Perceptual Image Patch Similarity (LPIPS) metrics

Table 1: LPIPS scores of in-the-wild data

SSv2	(1)	(2)	(3)	(4)	(5)
O_t	0.280	0.437	0.350	0.323	0.416
O_{t+1}	0.193	0.240	0.224	0.201	0.153
EGO4D	(6)	(7)	(8)	(9)	(10)
O_t	0.470	0.378	0.540	0.427	0.397
O_{t+1}	0.370	0.247	0.385	0.232	0.284

References

Zhang, Richard, et al. "The unreasonable effectiveness of deep features as a perceptual metric." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2018.